Project Task 03 - Submission Sheet

Software Testing - Graph Coverage

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Task 3 - Graph Coverage

a) Measure the coverage of your given project test suite (which includes the existing test suite as well as the tests that you created in previous project phases) by three graph coverage criteria which you can freely choose. Describe each individual result in 2-3 sentences.

Answer:

Answer:

The test suite is evaluated using **Node Coverage** and **Edge Coverage** as graph coverage criteria.

For **Node Coverage**, the tests achieve 57% coverage of the `trim_filename_if_needed` function. Key decision points are executed, including checks for filename length, trimming logic, and reconstruction with or without extensions. However, some deeper nodes, particularly those handling edge-case suffix logic, remain uncovered.

For **Edge Coverage**, the test cases explore multiple control flow paths, including both the no-trim path (N1 → N2 → N3 [True] → N4) and trim paths (N1 → N2 → N3 [False] → N5 → N6 and beyond). Additional tests cover edge cases such as empty strings, filenames without extensions, and long suffixes, allowing various decision branches to be exercised.

b) Extend the test suite with own tests, which have to fullfil **one** of the following criteria:

1) Increase the coverage values of all three coverage criteria that you applied in the previous subtask with at least 10 tests *per group member* (compare and describe the effects on coverage for each individual test)

Answer:

The test suite was extended with 8 additional tests designed to increase Node and Edge coverage of the `trim_filename_if_needed` function. These tests were focused on edge cases such as filenames with no extensions, long suffixes, empty strings, filenames with multiple dots, and exact-boundary lengths. As a result, the statement coverage of `downloads.py` increased from 54% to 57%, and the overall project coverage increased from 69% to 82%.

Each new test targets a specific control flow path or condition evaluation. For instance, one test verifies the behavior when only the extension fits within the allowed length, while another checks whether suffix preservation works with complex filenames. These cases help uncover additional branches and extend coverage of lines that were previously unexecuted, especially within the suffix trimming logic (lines 30–37).

The code is uploaded in the group Git repository. The file `test_trim_filename_if_needed_graph1.py` contains the newly added test cases that were written to improve graph coverage, particularly for edge cases such as filenames with no extension, long suffixes, empty input, and boundary length checks. In contrast, `test_trim_filename_if_needed_graph.py` includes the base test cases that cover the primary trim and no-trim logic paths. Only the final coverage results are included in this report; refer to the repository for full test code.

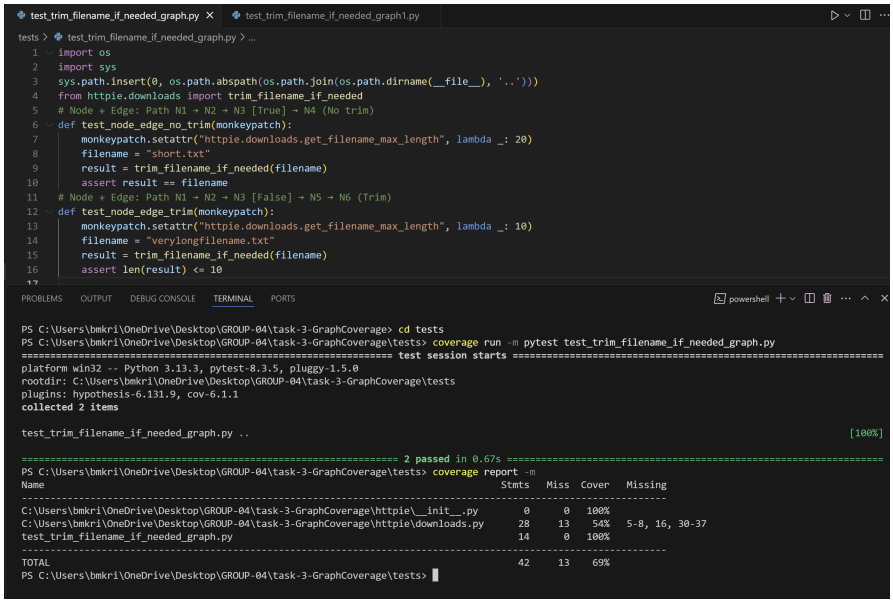

Figure 1: Coverage result for `test_trim_filename_if_needed_graph.py`

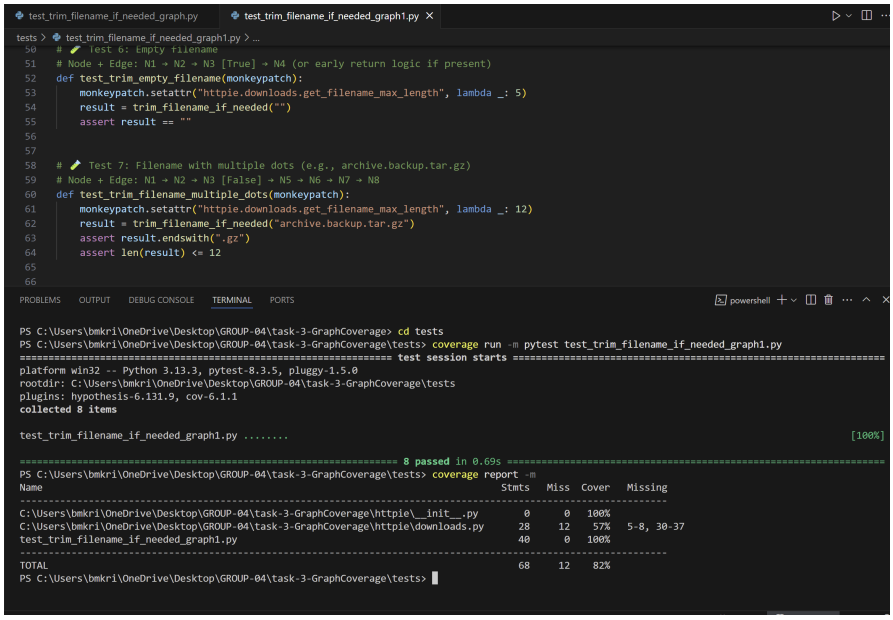

Figure 2: Coverage result for `test_trim_filename_if_needed_graph1.py`